SCIENCE / BC GRADE 8

Cells: structure & function

Compare plant and animal cells and connect parts to jobs.

Science 8

01 Cell theory

Living things consist of cells. Cells are the basic units of life, and new cells arise from existing cells. A model simplifies reality; it is not a photograph of every cell type.

02 Shared cell structures

Typical plant and animal cells have a cell membrane, cytoplasm, a nucleus, and mitochondria. The membrane regulates movement in and out; the nucleus contains most genetic material; mitochondria help release usable energy through cellular respiration.

03 Plant-cell features

Plant cells have a cell wall outside the membrane and often a large central vacuole. Photosynthetic plant cells have chloroplasts. Not all plant cells contain chloroplasts; underground root cells are a useful counterexample.

Worked example

Identify a cell from evidence

A cell has a cell wall, chloroplasts, and a large central vacuole. Together these observations support identifying it as a photosynthetic plant cell. A cell wall alone would not prove it is a plant cell because other groups also have walls.

Watch out: Plants also carry out cellular respiration. Photosynthesis does not replace all other energy processes.

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Try it. Then check it.

Work through these questions before looking at the answers.

1. Which structure regulates movement into and out of a cell?

2. Give two structures commonly used to distinguish a leaf cell from an animal cell.

3. Why is "all plant cells have chloroplasts" too broad?

Answer key

1. The cell membrane. A plant cell also has a wall, but the membrane regulates transport.

2. A cell wall and chloroplasts; a large central vacuole can provide additional evidence.

3. Some plant cells, such as many root cells, do not photosynthesize and lack chloroplasts.

One-minute recap

Close the notes. Explain one key idea in your own words, then retry the question you found most difficult.

About this guide

Original Study Bridge topic review prepared with AI assistance. Suggested course fit, not complete or official course coverage. Compare with your teacher's materials. Examples and practice questions are newly written.

Official course context: Science 8

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